

Project Check In

Team Name: BTripleJ



CIS 350 / June 2025

Presented by

Blake Collins

Joey Shotts

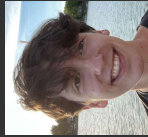
Josh Dobbs

Jason

Gray-Moore

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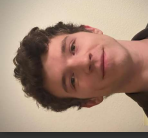
Blake Collins

CS major with EE background.
Have experience with: Microcontrollers
Python, and C. I worked with the microcontroller on this project



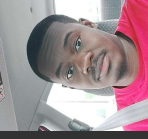
Joey Shotts

I'm a 4th year Computer Engineering Major. My fun fact is that I work in the Technology Showcase in the library. I have been working in Controls at JR Automation this past year.



Josh Dobbs

I'm a junior in Computer Science. I specialize in working in python and UI. I develop the main GUI of this project.



Jason Gray-Moore

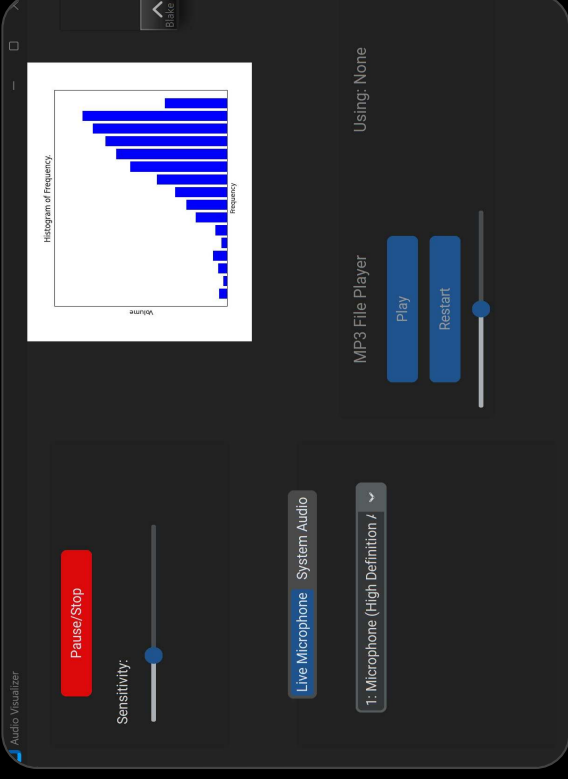
I am Computer Engineering Major. I write software for Savant Automation AGVS. I worked with the microcontroller for this project.



Arduino Audio Visualizer with Python GUI

We used Custom Tkinter for this project. It allowed for simple grid system GUI creation. Widgets are easily added to a grid in which you can interact with them to adjust how the LED will be displayed. The GUI is simple and easy to use:

- Play and Pause and Sensitivity in the top left for controls directly to the output
- Live Mic and System Audio options for where the input comes from
- An MP3 player for selecting mp3 files to use
- And a graph visualizer to see real input



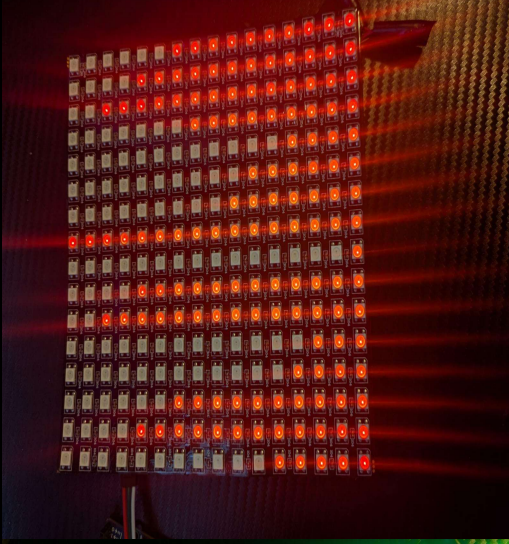
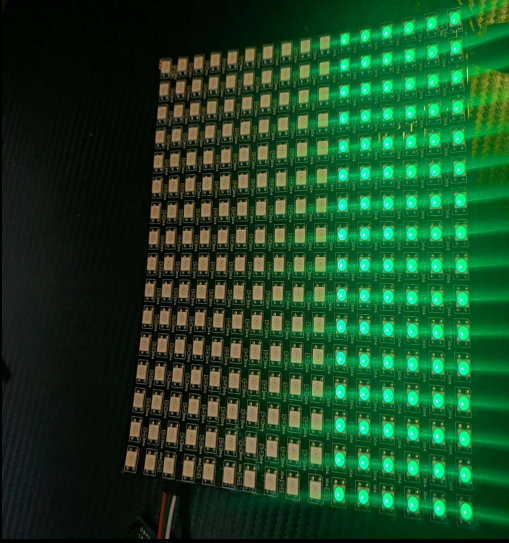
Arduino Display

After the audio or microphone input is analyzed, the data is transferred over serial to the arduino. We use the FastLED library to be handled setting each LED according.

- Use CHSV to determine color for LED
- Count the number of commas to know if Amplitude or Frequency

110,40

0,10,12,14,9,4,6,7,16,1,12,14,3,4,10,12,5



How it works



Custom Tkinter

This is the library we used to create the python GUI. It is based off of the popular Tkinter Library and is a simple way to create a functional GUI.



Audio Processing

We are using the library sounddevice to stream audio from a variety of sources. We then are using numpy to process the data in real time gathering the max volume and a collection of frequency magnitudes.



Comms

Sending data over Serial to the Arduino. We are using commas to separate each input, and a new line indicates that the input is complete. First input would be the Color using CHSV followed by the amplitude or the frequency.

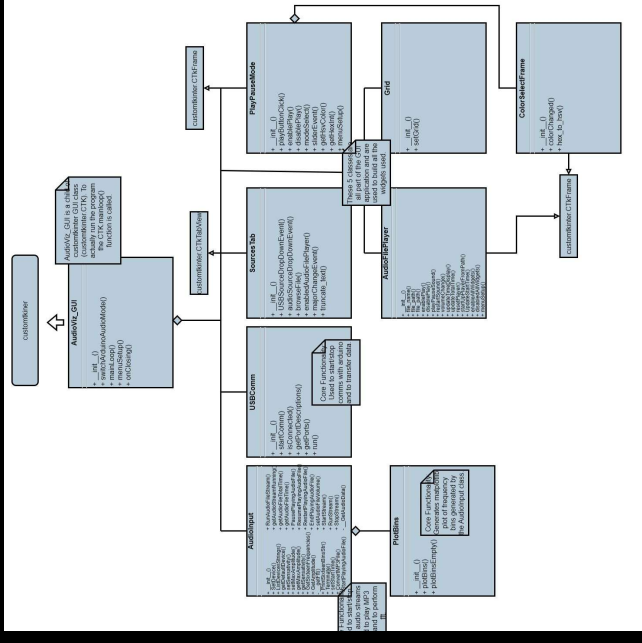


Arduino

parse data from Serial and determine what mode and color to set the LEDs. With FastLED library we display frequency or amplitude of the audio.

What has changed since the mid checkpoint?

- We used trello more effectively
- We updated our functional requirements
- We built a better UML diagram
- We got MP3 audio files playing
- We transferred color along with amplitude and frequency information
- We tweaked some bugs in the program



Project Timeline

We are using a GANTT chart to keep track of our project. So far we have been keeping up with our timeline.

[illegible]

We Mostly Hit Our Time Goals



Challenges with Software Engineering

- Using Trello: we could have used it earlier in the project to be better organized
- Github as we had to figure out what branches we were working off of.
- Creating Software Requirements
- Completing the technical details of the project
- Scaling issues when sending data to the arduino

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Look at our sick demo...

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Thank you

Ready for what's next?

Let's talk

Any Questions?

B Triple J

Josh, Joey, Blake, Jason